Eco-Friendly Management of Insect Pests Infesting Cucumber (*Cucumis sativus* L.)

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INTRODUCTION

Cucumber (*Cucumis sativus* L.) is a widely cultivated vegetable belonging to the gourd family (Cucurbitaceae) and it is originated in India. Known for its crisp texture and refreshing taste, cucumber is a popular ingredient in salads, sandwiches, and various culinary dishes across the globe. Originating in South Asia, cucumbers have been cultivated for thousands of years, with evidence of their use dating back to ancient civilizations in Egypt and India. In addition to their culinary uses, cucumbers are valued for their nutritional benefits. They are low in calories, high in water content, and provide essential vitamins and minerals, including vitamin K, vitamin C, and potassium (Ali & Geng, 2019). Furthermore, cucumbers contain antioxidants that can help promote health and well-being. Indigenous to India, cucumbers exhibit considerable variation in morphological characteristics, such as growth habit, fruit size, and color. Continued selection has led to the accumulation of numerous landraces with limited local distribution across different growing regions.

Cucumbers thrive in cool climates and require well-drained, nutrient-rich soil for optimal growth. They can be grown in various environments, including open fields, greenhouses, and hydroponic systems, making them versatile for both commercial production and home gardening. The crop is typically grown as a winter to summer annual, taking approximately 50 to 70 days from planting to harvest, depending on the variety.

In 2022, India produced approximately 1,363,430 tonnes of cucumbers in 20 thousand hectare land. West Bengal led the production with about 326,820 tonnes, contributing 20.32% of the total. Other significant producing states included Madhya Pradesh (14.76%), Haryana (11.38%), and Uttar Pradesh, which accounted for around 103,740 tonnes, or 6.45% of the total production (Ministry of Agriculture Govt of India 2023). In 2022, Uttar Pradesh produced approximately 103,740 tonnes of cucumbers, contributing about 6.45% to India's overall cucumber production. This positions the state as one of the significant producers of cucumber in the country. Major producing states included Uttar Pradesh, West Bengal, Punjab, and Haryana, where favorable growing conditions supported high yields (NHB, 2022)

This production plays a significant role in meeting domestic demand for fresh vegetables and contributes to the agricultural economy. However, repeated cultivation has heightened pest and disease problems, which are key factors contributing to low yields. Among various pest's aphid, thrips, mite and leaf minor is a common pest of cucumbers, causing significant losses in both quality and yield. These sucking pest damage plants by rasping the leaves, resulting in yellow, white, or brown discoloration, crinkling, and even death. Heavily infested fields may take on a bronze hue. Cucumber beetle's pests, particularly the striped and spotted varieties, can damage plants by feeding on leaves and transmitting bacterial wilt.

Cucumber, like other members of the cucurbit family, is susceptible to various insect pests from germination to harvest. Among these, the red pumpkin beetle (*Raphidopalpa foveicollis*), fruit flies (*Bactrocera cucurbitae*), and hadda beetle (*Epilachna dodecasigma*) are particularly concerning (Gupta, 2004). Both *B. cucurbitae* and *R. foveicollis* can inflict complete fruit damage, especially in pumpkins, cucumbers, and bitter gourds (Jyoti and Rajbhandari, 2015). Yield losses attributed to *B. cucurbitae* vary, with cucumbers experiencing 20-39% losses, while bitter gourd can suffer losses of 60-80%, ivy gourd and snake gourd around 63%, and sponge gourd about 50%. In addition to causing direct harm, many of these pests also serve as vectors for various viral diseases. This study aims to investigate the population dynamics of *Amrasca biguttula biguttula*, *Bactrocera cucurbitae*, *Liriomyza trifolii*, and *Bemisia tabaci* in cucumber cultivation.

The list of components includes various substances commonly used in agriculture, particularly for pest management and plant health. Neem oil, derived from the seeds of the neem tree, is a natural pesticide with a concentration of 5000 ppm, known for its effectiveness in repelling and disrupting the life cycle of pests. *Verticillium lecanii* is a beneficial fungus that acts as a biocontrol agent, targeting and infecting various insect pests, thus reducing their populations. *Metarhizium anisopliae* is another entomopathogenic fungus that is utilized for its ability to infect and kill insects, making it an important tool in integrated pest management.

Fish oil rosin soap serves as an insecticidal soap that helps control soft-bodied pests by suffocating them and disrupting their cell membranes. Primarily used as an insecticidal soap in agriculture and horticulture to control soft-bodied pests such as aphids, mites, and whiteflies. It can also act as a surfactant, helping other pesticides or herbicides stick to plant surfaces. It disrupts the cell membranes of pests, leading to dehydration and death.

Mineral oil is often used as a horticultural oil to smother pests and reduce disease incidence by forming a protective layer on plant surfaces. Mineral oil is used as a pesticide, particularly as a horticultural oil. It controls pests by suffocating them, especially insect eggs, scales, and mites on plants. It is also used to prevent fungal diseases.

Polyether modified trisiloxane is a type of silicone-based surfactant that is used to enhance the effectiveness of agricultural chemicals and surfactant that enhances the spreading and penetration of pesticides, improving their efficacy. It creates a thin layer on the insect or egg surface, blocking their breathing pores (spiracles) and preventing respiration. A type of silicone-based surfactant that is used to enhance the effectiveness of agricultural chemicals.Commonly used as a spreader and wetting agent in pesticide formulations. It helps reduce the surface tension of water, allowing pesticides, herbicides, or fertilizers to spread more evenly on plant surfaces and penetrate better into tissues.

Finally, spinosad, derived from naturally occurring bacteria, is a potent insecticide effective against a wide range of pests, offering a less harmful alternative to synthetic chemicals. Together, these components represent a combination of natural and modified substances aimed at promoting effective pest management while minimizing environmental impact. Spinosad is effective against a variety of insect pests such as caterpillars, thrips, leafminers, and fruit flies. It is commonly used in organic farming, as well as in conventional agriculture. Spinosad works by affecting the nervous system of insects, leading to paralysis and death.

Verticillium lecanii, now more commonly known as *Lecanicillium lecanii*, is a naturally occurring fungus that acts as an entomopathogen, meaning it infects and kills insects. It is widely used as a biological control agent in agriculture and horticulture to manage pest populations, particularly soft-bodied insects. It is primarily effective against aphids, whiteflies, mealybugs, and thrips and it can also control some scale insects. The mode of action of fungus infects pests by attaching to their cuticle (outer body surface) and penetrating it and once inside, it grows and proliferates, ultimately killing the insect by consuming its internal contents. Infected insects usually die within a few days, and the fungus produces spores that can spread to other pests.

Metarhizium anisopliae is a naturally occurring entomopathogenic fungus widely used in agriculture as a biological control agent to manage various insect pests. It infects and kills a broad range of insect species, making it an essential component of integrated pest management (IPM) strategies. Effective against a wide variety of pests, including termites, locusts, beetles, weevils, aphids, and caterpillars. Particularly useful for controlling soil-dwelling insects such as root grubs and termites. The fungus infects insects by adhering to their cuticle (outer surface) and penetrates the cuticle and multiplies inside the insect, causing death through nutrient depletion and mechanical damage. The infected insect eventually dies within a few days, and the fungus releases spores that can spread and infect other insects. To address the challenges posed by chemical insecticides, this investigation aims to evaluate Eco-Friendly Management of Insect Pests Infesting Cucumber (*Cucumis sativus* L.).

The objectives of the study are as follows:

1. To evaluate eco-friendly approaches against major sucking insect pests of cucumber.
2. To evaluate eco-friendly approaches against major defoliator insect pests of cucumber
3. To assess the impact of treatments on yield loss.
4. To calculate the economics of the treatments

BRIEF REVIEW OF LITERATURE

1. To evaluate eco-friendly approaches against major sucking insect pests of cucumber

Andrew Beattie (2005) highlighted the importance of petroleum-based spray oils in the pest management of citrus crops, especially as part of integrated pest management (IPM) strategies. He suggested applying horticultural mineral oil at concentrations ranging from 0.25% to 0.5% to effectively combat citrus greenhouse thrips.

Bhatnagar (2009) noted that, ten days after the initial spray, the imidacloprid treatment resulted in the lowest thrip population, followed by acetamiprid, thiamethoxam, and dimethoate. Among natural alternatives, neem seed kernel extract (NSKE) exhibited the lowest thrip count, whereas the other treatments did not show a significant difference compared to the untreated control.

Egho and Emosairue (2010) observed that mineral oils have proven effective in managing insect pests across different crops. Their research assessed the efficacy of three types of mineral oils—premium motor spirit (PMS), dual purpose kerosene (DPK), and automotive gas oil (AGO)—at a concentration of 4% for controlling the legume bud thrips, *Megalurothrips sjosstedti*.

Janaki et al. (2011) carried out field experiments to evaluate the effectiveness of various biopesticides, including *Beauveria bassiana*, *Pseudomonas fluorescens*, Spinosad, and fish oil rosin soap, in controlling the papaya mealybug. Of the biopesticides assessed, fish oil rosin soap showed the most significant efficacy, resulting in a 51.99% reduction in mealybug populations.

2. To evaluate eco-friendly approaches against major defoliator insect pests of cucumber

Azam (1991) conducted a study to evaluate the toxicity of neem oil at varying concentrations (0.5%, 0.75%, 1.0%, 1.25%, and 1.5%) on the larvae of *Liriomyza trifolii* (leaf miner) on cucumber leaves. The study revealed that more than 60% mortality of larvae and pupae occurred when neem oil was applied at concentrations between 1.0% and 1.25%. However, concentrations lower than 1% were less effective in controlling the pests, leading to minimal mortality. Interestingly, increasing the neem oil concentration to 1.5% did not result in a significant increase in mortality, suggesting that the optimal concentration for controlling Liriomyza trifolii larvae lies between 1.0% and 1.25%, with diminishing returns at higher concentrations in the laboratory setting.

Botin et al. (2004) conducted a study to examine the effects of aqueous neem extract (5%) and neem oil (4.5%) on the oviposition behavior of the Mexican fruit fly (*Anastrepha ludens*) in Valencia orange under controlled conditions. The results demonstrated that both treatments were highly effective in repelling the flies, with complete oviposition repellency observed in each case. This suggests that neem-based products can serve as a potent deterrent to the Mexican fruit fly, preventing egg-laying on Valencia oranges and potentially providing a natural method for managing this pest.

Silva *et al.* (2011) studied the effects of biopesticides, including *Metarhizium anisopliae*, *Verticillium lecanii*, and *Beauveria bassiana*, on thrips in potato crops. The application of these biopesticides significantly reduced the population of *Thrips palmi* compared to untreated controls, with the highest mortality percentages achieved by *M. anisopliae* at 1.7×10131.7 \times 10^{13}1.7×1013 conidia/ha and *V. lecanii* at 1.5×10131.5 \times 10^{13}1.5×1013 conidia/ha. While the number and weight of tubers showed no significant differences among treated variants, there was a positive trend compared to the control.

Gawande *et al.* (2012) investigated the efficacy of the entomopathogenic fungus *Verticillium lecanii* against whitefly nymphs (*Bemisia tabaci*) in vitro. The study involved exposing the nymphal population to various concentrations of the bioagent, specifically from 10910^9109 to 10610^6106 spores/ml, both alone and in combination with different oils—Neem oil, D-C-Tron plus oil, and liquid paraffin oil—at a concentration of 1%. Each treatment was replicated three times. The researchers measured dose mortality (LC50) and time mortality (LT50) to evaluate the whiteflies' response to the treatments. Among all the treatments tested, Dimethoate at 0.05% proved to be the most effective. When it came to the application of the bioagent, *V. lecanii* at 10910^9109 spores/ml combined with Neem oil and D-C-Tron plus oil at 1% concentration showed promising results, achieving lower LT50 values and higher mortality rates in the test population.

Gabrielet al. (2014) assessed various non-ionic surfactants to enhance the insecticidal performance of bio-based products and agrochemicals against *Bemisia tabaci* biotype B whiteflies. They found that trisiloxane-based surfactants, particularly Break-thru and Silwet L-77, showed the highest toxicity to both early and late instar nymphs, along with superior wetting properties. The study also explored whether sub-lethal concentrations of these surfactants could boost the efficacy of fungal entomopathogens *Beauveria bassiana* and *Isaria fumosorosea*. Results indicated that these surfactants did not adversely affect conidial germination and significantly increased nymph mortality, with mostly additive and synergistic effects observed. In screenhouse trials, mixtures of the fungi with Silwet L-77 at 200 ppm achieved 72-74% mortality in early nymphs, highlighting the potential of silicon-based surfactants in integrated whitefly management.

Chaudhari et al. (2017) conducted a trial on cucumber under shade net conditions to assess the efficacy of six bio-rational components in managing Thrips tabaci (Lindeman), with a standard check of spinosad at 1.2 ml/l. The study found that *Verticillium lecanii* at 4 g/l (average 59.50%) and trisiloxane at 0.25 ml/l (average 57.84%) were the most significant eco-friendly options for suppressing thrips. These were followed by *Metarhizium anisopliae* at 4 g/l (average 57.20%) and neem oil at 1 ml/l (average 56.47%). Mineral oil at 0.5 ml/l (average 54.84%) and fish oil rosin soap at 1 ml/l (average 54.11%) were also effective, ranking next among the bio-rational components.

Hossain *et al.* (2018) investigated the foraging behavior of **Apis mellifera**, noting that bees spent significantly more time per flower during morning hours (seconds per flower) and foraged fewer flowers (7.9 flowers per minute) compared to evening hours. The study found a higher number of nectar foragers (6.03 per m^2 per 10 minutes) than pollen foragers (5.16 per m^2 per 10 minutes). Most pollen foragers were active in the morning (6.59 per m^2 per 10 minutes), while nectar foragers peaked during noon (6.63 per m^2 per 10 minutes). Hand pollination yielded the highest fruit set at 70.68%, whereas the percentage of misshapen fruits was greatest without honey bee pollination (24.35%). The absence of honey bee pollination led to the lowest percentage of healthy fruits (75.25%). Hand pollination outperformed the other pollination methods in terms of total fruit weight (985.13 g), number of seeds per fruit (425.22), fruit diameter (27.1 cm), fruit length (26.7 cm), and weight of 1000 seeds (28.64 g).

Billur et al. (2022) conducted research at the Agronomy Farm, College of Agriculture, Dapoli, during the Kharif season of 2016, examining seven flowering plants—cosmos, cowpea, marigold, mustard, niger, sesamum, and zinnia—planted along the borders of cucumber plots to conserve natural enemies. The study found that niger, sesamum, mustard, cowpea, and marigold effectively reduced the incidence of red pumpkin beetles. Additionally, niger, mustard, marigold, and cowpea were the most effective in mitigating flea beetle populations, while niger, marigold, cowpea, mustard, and sesamum were best for controlling serpentine leaf miners. Furthermore, niger, cosmos, cowpea, sesamum, and zinnia significantly reduced fruit fly infestations. The findings of this study contribute to eco-friendly pest management strategies for cucurbits.

3. To assess the impact of treatments on yield loss.

Mandal *et al.* (2006) conducted experiments over two consecutive summer seasons in 2000 and 2001 to assess the effectiveness of various neem-based integrated management strategies against the okra aphid, *Aphis gossypii*. The treatments involved applying neem cake to the soil at 200 kg/ha, along with three foliar applications of neem seed kernel extract (5%), neem oil (3%), Amrutguard (0.5%), and neem leaf decoction (0.5 kg). The findings revealed that the neem seed kernel extract (5%) was the most effective treatment, significantly reducing aphid populations and increasing the yield of marketable fruit. Moreover, these integrated approaches were found to be profitable, resulting in the highest cost-benefit ratio among the treatments.

Thangavel et al. (2011) conducted a field experiment at the Agricultural College and Research Institute in Madurai from June to November 2009 to evaluate the effectiveness of botanicals and fish oil rosin soap in managing major pests of coleus. Over the course of more than ten applications at fortnightly intervals, they monitored the incidence of thrips, scale insects, and defoliators. The results indicated that Azadirachtin 0.15% EC (Neem Gold) effectively reduced thrips and defoliator damage, with average leaf damage recorded at 6.1% and 6.6%, respectively. Fish oil rosin soap (25 g/l) was particularly effective against scale insects, achieving the lowest mean leaf damage of 4.8%. Notably, Azadirachtin also resulted in the highest yield, producing 20,836 kg of wet tubers per hectare, in contrast to just 12,451 kg in the untreated control group.

4. To calculate the economics of the treatments

Mandal et al. (2006) evaluated neem-based integrated management strategies against the okra aphid, *Aphis gossypii*, over two summer seasons in 2000 and 2001. They tested treatments including neem cake (200 kg/ha) and three foliar applications of neem seed kernel extract (5%), neem oil (3%), Amrutguard (0.5%), and neem leaf decoction (0.5 kg). The neem seed kernel extract was found to be the most effective, significantly reducing aphid populations and increasing marketable fruit yield. These integrated approaches also demonstrated a high cost-benefit ratio, making them profitable.

Pawar et al. (2019) evaluated various pest management modules for controlling major pests of cucumber (*Cucumis sativus L.*). Their study found that the Integrated Pest Management (IPM) module effectively reduced populations of thrips (4.31 per three leaves), whiteflies (3.80), and aphids (3.90). Additionally, the IPM module showed significant superiority against leaf miners and fruit flies, resulting in 21.15% damaged leaves and 10.84% damaged fruits. The highest cucumber yield of 230 q/ha was recorded with the IPM module, which also achieved the best Incremental Cost-Benefit Ratio (ICBR) of 1:13.86, compared to 160 q/ha in the untreated control.

METHODOLOGY

The study will be undertaken in experimental farm department of Agricultural Entomology, Janta Mahavidyalaya Ajeetmal Auraiya, UP. The details of experiment are given below

1. Crop	: Cucumber
2. Variety	: Pusa Long green
3. Design	: Randomized Bock Design
4. Replication	: 03
5. Treatments	: 08
6. Plot to plot distance	: 0.5cm
7. Replication to replication distance	: 1m
8. Plot size	: 4 m x 3 m
9. Spacing	: 2.0 m x 01 m
10. Season	: Zaid Season 2024-25 and 2025-26
11. Date of Sowing	: ………………. (First year)
12. Date of Sowing	: ………………... (Second year)
13. Seed rate	: 2.5 kg/ha

14. RDF : 100:75:75 NPK kg/ha +FYM 20 ton/ha

Table: 01. Eco-friendly management of insect pest with following treatment's

S. No.	Notation	Treatment's	Dose
1.	T_1	Neem oil 5000ppm	1ml/litre
2.	T_2	*Verticillium lecanii*	4g/litre
3.	T_3	*Metarhizium anisopliae*	4g/litre
4.	T_4	Fish oil rosin soap	1ml/litre
5.	T_5	Mineral oil	0.5ml/litre
6.	T_6	Polyether modified trisiloxane	0.25ml/litre
7.	T_7	Spinosad	1.2ml/litre
8.	T_8	Untreated control	-

Note:- Neem oil is prepared by first harvesting mature neem seeds and cleaning them to remove dirt and foreign material then complete dry. The oil is then extracted, typically through cold pressing, where the seeds are mechanically pressed to release the oil, preserving its beneficial compounds. After extraction, the oil is filtered to remove any remaining seed particles and then stored in airtight containers, away from direct sunlight, to maintain its potency.

Spray schedule:
1. First spray will be given at the initiation of insect infestation.
2. Second spray at 15 days after first spray

METHODS OF TAKING OBSERVATIONS

Sucking insect pests

For recording observations on sucking pest viz., aphid, white fly, thrips, and spider mite for the observation of sucking pests i.e. white flies, jassid and thrips will be recorded in three leaves, upper, middle and bottom leaves of the plant on randomly selected ten plants. Then, the mean number of sucking pests per three leaves per plant will calculated.

After a set period post-treatment (e.g.,1,3,5, 7, 14, days), assess the efficacy of biopesticides based on reducing pest population and compare pest counts in treated plots with control plots.

Defoliator insect pests

For the observation of defoliators i.e. cucumber beetles, fruit fly and leaf minor, the number of larvae or caterpillars will be counted through direct visual counting method on randomly selected five plants or by shaking the plants. Observations will be recorded at weekly interval at randomly selected plants leaving border rows and after that the mean number of larvae per meter row will calculated.

After a set period post-treatment (e.g.,1,3,5, 7, 14, days), assess the efficacy of biopesticides based on reducing defoliators pest population and compare pest counts in treated plots with control plots.

Fruit Damaging insect pests

The major fruit damaging insect pest of cucumber is fruit fly (*Bactrocera cucurbitae*) and observations will be recorded at weekly interval at randomly selected plants leaving border rows and after that the mean number of fruit fly per meter row will be calculated.

Yellow Sticky Trap

Yellow Sticky Trap were used to monitor insect Population in the cucumber field. The yellow Sticky Trap of 20 ×15cm sized was hung and was adjusted vertically with the help of a wooden stick/pole

The trap were hung at three different heights (50,100 and 150cm) for the respective treatment of the plot, The Yellow traps were replaced by fresh ones at an interval of 3 days. Each Yellow trap was placed in the monitoring and Eco-friendly management of fruitfly

Middle of the randomly selected plots three replications were maintained for each treatment. The traps were checked regularly and were counted the number of fruit flies captured every 3 days interval. The traps were maintained in the field from the flower initiation stage to the last harvest covering the entire reproductive stage of Cucumber

Expected Outcomes

1. A reduction in pest populations and associated crop damage under eco-friendly management strategies.
2. Improved cucumber yield and fruit quality through sustainable pest management practices.

3. Evidence that eco-friendly methods are viable alternatives to chemical pesticides, contributing to long-term pest control and environmental protection.
4. Recommendations for integrated pest management (IPM) strategies for cucumber cultivation.

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